

7.20 (a) For a single frequency, the Maxwell equation in 1D is $\left(\frac{\partial^2}{\partial x^2} + \frac{n(\omega)^2}{c^2} \omega^2\right) u(\omega, x) = 0$

The general solution to this equation is $u(x, \omega) = A e^{i n(\omega) \omega x / c} + B e^{-i n(\omega) \omega x / c}$.

Then, the general solution in time space can be constructed by Fourier transform as

$$u(x, t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} d\omega e^{-i\omega t} \left[A(\omega) e^{i n(\omega) \omega x / c} + B(\omega) e^{-i n(\omega) \omega x / c} \right]$$

(b) Take the complex conjugate.

$$\begin{aligned} u^*(x, t) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} d\omega e^{i\omega t} \left[A^*(\omega) e^{-i n^*(\omega) \omega x / c} + B^*(\omega) e^{i n^*(\omega) \omega x / c} \right] \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} d\omega e^{-i\omega t} \left[A^*(-\omega) e^{i n^*(-\omega) \omega x / c} + B^*(-\omega) e^{-i n^*(-\omega) \omega x / c} \right] \end{aligned}$$

If $u(x, t)$ is real, $u^*(x, t) = u(x, t)$, we must have

$$A^*(-\omega) = A(\omega), \quad B^*(-\omega) = B(\omega), \quad n^*(-\omega) = n(\omega), \quad \text{or } n(-\omega) = n^*(\omega).$$

(c) At $x=0$, $u(0, t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} e^{-i\omega t} [A(\omega) + B(\omega)] d\omega$

and $\frac{\partial}{\partial x} u(0, t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} e^{-i\omega t} \frac{i n(\omega) \omega}{c} [A(\omega) - B(\omega)] d\omega$

Then, $A(\omega) + B(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} u(0, t) e^{i\omega t} dt$

$$A(\omega) - B(\omega) = -\frac{i c}{n(\omega) \omega} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} \frac{\partial}{\partial x} u(0, t) e^{i\omega t} dt.$$

Therefore, $\begin{Bmatrix} A(\omega) \\ B(\omega) \end{Bmatrix} = \frac{1}{2} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} dt e^{i\omega t} \left[u(0, t) \mp \frac{i c}{n(\omega) \omega} \frac{\partial}{\partial x} u(0, t) \right]$